BEINING BAO

PhD Student, AI of Health & Life Science (Biostatistics) | City University of Hong Kong
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RESEARCH INTERESTS

AI for health and life sciences; medical large language models; privacy-preserving EHR transformation and clinical text generation; medical knowledge graphs; large-scale recommendation and language models; AI safety and reproducible machine learning.

EDUCATION

Ph.D. in AI of Health & Life Science (Biostatistics)
City University of Hong Kong, Hong Kong | Sept 2026 - Jun 2030 (expected)
Status: Current PhD student. Advisor: Prof. Jun Yan. Collaborators/advisors: Prof. Maolin Wang, Senior Engineer Yao Wang. Prior mentor: Prof. Xiangyu Zhao (Department of Data Science).

M.Sc. in Data Science
City University of Hong Kong, Hong Kong | Aug 2025 - Jun 2026
Graduated with distinction (First Class Honours). Research training in data science, AI for Science, medical AI, and reproducible machine learning.

B.Sc. in Business Studies (Double Degree)
Lancaster University, UK / Beijing Jiaotong University, China | Sept 2020 - Jun 2025
Undergraduate training in analytics, business research, and data-driven decision making.

RESEARCH / WORK EXPERIENCE

Research Assistant, AI for Science
HKAI-Sci, City University of Hong Kong | Jan 2026 - Present
Research on medical LLMs, privacy-preserving clinical data transformation, medical knowledge graphs, and AI safety; model implementation, experiments, paper writing, and reproducible research code. Supervisor: Prof. Jun Yan.

Large Language Model Intern
Tencent (WeChat Search / Hunyuan), Beijing | Dec 2024 - Mar 2025
Text preprocessing for medical/legal documents; prompt and synthetic-data pipelines; improved retrieval efficiency and labeling consistency.

Data Operations Intern
Joy Media, Hangzhou | Jul 2023 - Sept 2023
KOL live-stream and short-video analytics; competitive analysis to guide content strategy and operations.

Data Research Intern
Beijing International Big Data Exchange Co., Ltd. | Aug 2022 - Sept 2022
Industry and competitor research; data-driven reports for product planning and marketing strategy.

SELECTED PUBLICATIONS

* marks myself. Selected accepted / published works:

• **GRAND: A Robust Diffusion Framework for Multi-Granularity Graph Anomaly Detection in Web Platforms.** Maolin Wang, **Beining Bao***, Hongyu Chen, Zichun Liu, Lang Fu, Jun Chu, Langzhang Liang, Zenglin Xu. TheWebConf 2026 (Web4Good), CCF-A. DOI: 10.1145/3774904.3793012.

• **A Serial Two-Stage Framework for Robust Multimodal Fake News Detection via Adaptive Reasoning.** Maolin Wang, Ziting Mai, Zichun Liu, **Beining Bao***, Hongyu Chen, et al. ACM SIGKDD 2026, CCF-A.

• **Rethinking Semantic-Collaborative Integration: Why Alignment Is Not Enough.** Maolin Wang, Dongze Wu, Jianing Zhou, Hongyu Chen, **Beining Bao***, Yu Jiang, et al. ACM SIGIR 2026 (Perspectives), CCF-A. DOI: 10.1145/3805712.3808539.

• **MuSeR: Scalable Long-sequence Recommendation with Multi-interest Modeling.** Yongkang Fu, **Beining Bao***, Yu Jiang, Xiangyu Zhao, et al., Maolin Wang. IEEE ICDM 2026 (Applied Track), CCF-B. Deployed at Baidu. arXiv:2609.23677.

• **Deep Tensor Factorization for Modeling Non-Uniform Temporal Dynamics with Adaptive Temporal Smoothing.** Hongyu Chen, **Beining Bao*** (co-first), Zichun Liu, Tianshuo Wei, Maolin Wang, Xiaopeng Li, Hing Cheung So. IEEE SAM 2026.

• **Privacy-Preserving EHR Data Transformation via Geometric Operators: A Human-AI Co-Design Technical Report.** Maolin Wang†, **Beining Bao***†, et al., Jun Yan. arXiv:2603.22954, 2026.

• **Housing Rental Information Management and Prediction System Based on CatBoost Algorithm - A Case Study of Halifax Region.** Shuangrun Shao, Bingxi Zhao, Xiangen Cui, Yihong Dai, **Beining Bao***. IJCRS 2024 / LNCS, CCF-B.

Selected manuscript under review: From Impulse to Interest: Curiosity-Aware Click Modeling for Long-Term Preference Estimation. Chengxi Wei, **Beining Bao***, Hongyu Chen, et al. ACM RecSys 2026 (Main Track), under review.

SELECTED PROJECTS & HONOURS

• Intelligent Elderly Care Matching Platform (YangLaoTong), Nov 2024 - May 2025: fuzzy matching, TF-IDF, CatBoost over 26,000+ nursing homes; IBM Watson dialogue; 2,086 questionnaires with NLP refinement.

• Honours: Third place, IJCRS International Data Mining Competition Final (2024); Best Project Award, MSc in Data Science (2026); Excellence Award, National College Student Data Analysis Popular Science Knowledge Competition (Oct 2023).

SKILLS

Python, PyTorch, scikit-learn, CatBoost, Transformers, Graph ML, Medical LLMs, Medical knowledge graphs, AI safety, LLM prompting & data generation, reproducible pipelines. Languages: English, Mandarin.